

Supply Chain Management

Lecture 3

MM BASC 523

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Wriston Manufacturing Corporation

Decisions

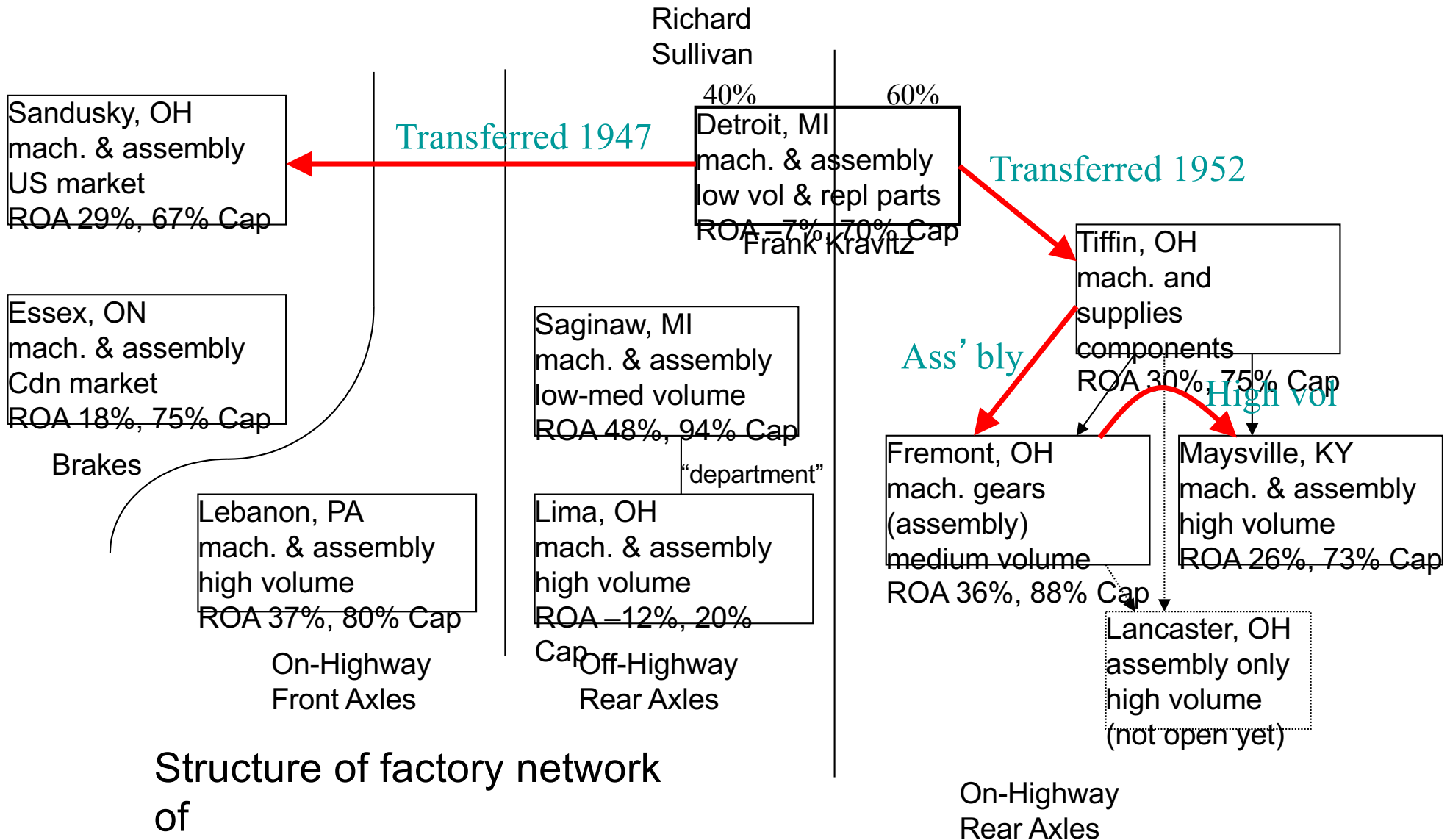
- What is the decision to be made?
- How did you decide, what framework did you use?

Operational Decisions...

- **Plant?**

- **Products?**

Picture of factory network



Detroit Plant

- From a process perspective, what type of a plant is Detroit?

Types of processes

Job Shops and Flow shops

**High Variance in
inputs and tasks**

**Lower volume of a
specific type**

**Likely not suitable
for assembly line
approach**

**Higher Per unit cost
of product/procedure**

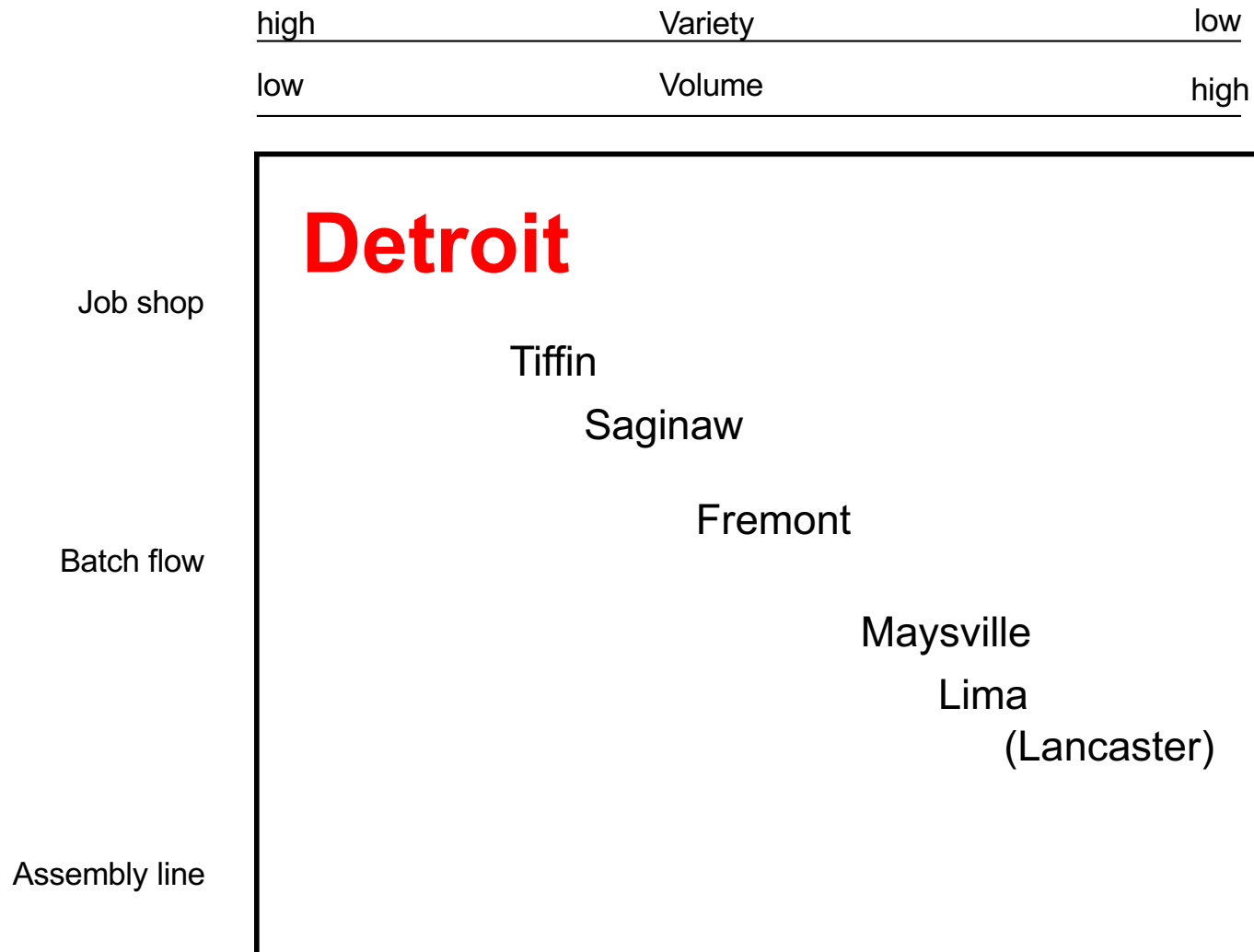
**Lower Variance in
inputs and tasks**

**Higher volume of a
specific type**

**suitable for
assembly line
approach**

**Lower Per unit cost
of product/procedure**

A Product-Process Matrix for Wriston



Ratios from Exhibit 2A

<i>Plant</i>	<i>(new) Empl. Per M\$ Sales</i>	<i>Variable burden rate¹</i>	<i>Total burden rate²</i>	<i>Number of product families³</i>	<i>Number of product models³</i>
Sandusky, Ohio	6.99	1.11	3.58	5	10
Essex, Canada	8.72	1.64	5.30	4	8
Detroit, Michigan	10.11	2.13	6.00	20	120
Saginaw, Michigan	8.05	1.35	4.10	10	110
Lima, Ohio	10.33	1.56	5.05	4	80
Lebanon, Pennsylvania	6.89	0.85	2.64	2	30
Tiffin, Ohio	6.05	1.12	3.50	6	60
Fremont, Ohio	8.01	1.20	3.65	10	40
Maysville, Kentucky	7.34	0.71	2.35	2	20

So What?

Insights from Practice!

(1) Focused factory

(2) FUND,
MANAGE AND
MEASURE differently

Focused Factory

- Innovation Versus Mass High Volume Production
- High touch customized service Versus Low volume undifferentiated service.

Responsibility Accounting

- Costs & revenues measured, allocated based on the role of a division or facility.
- Integrates Operations, finance and strategy in producing internal financial statements.

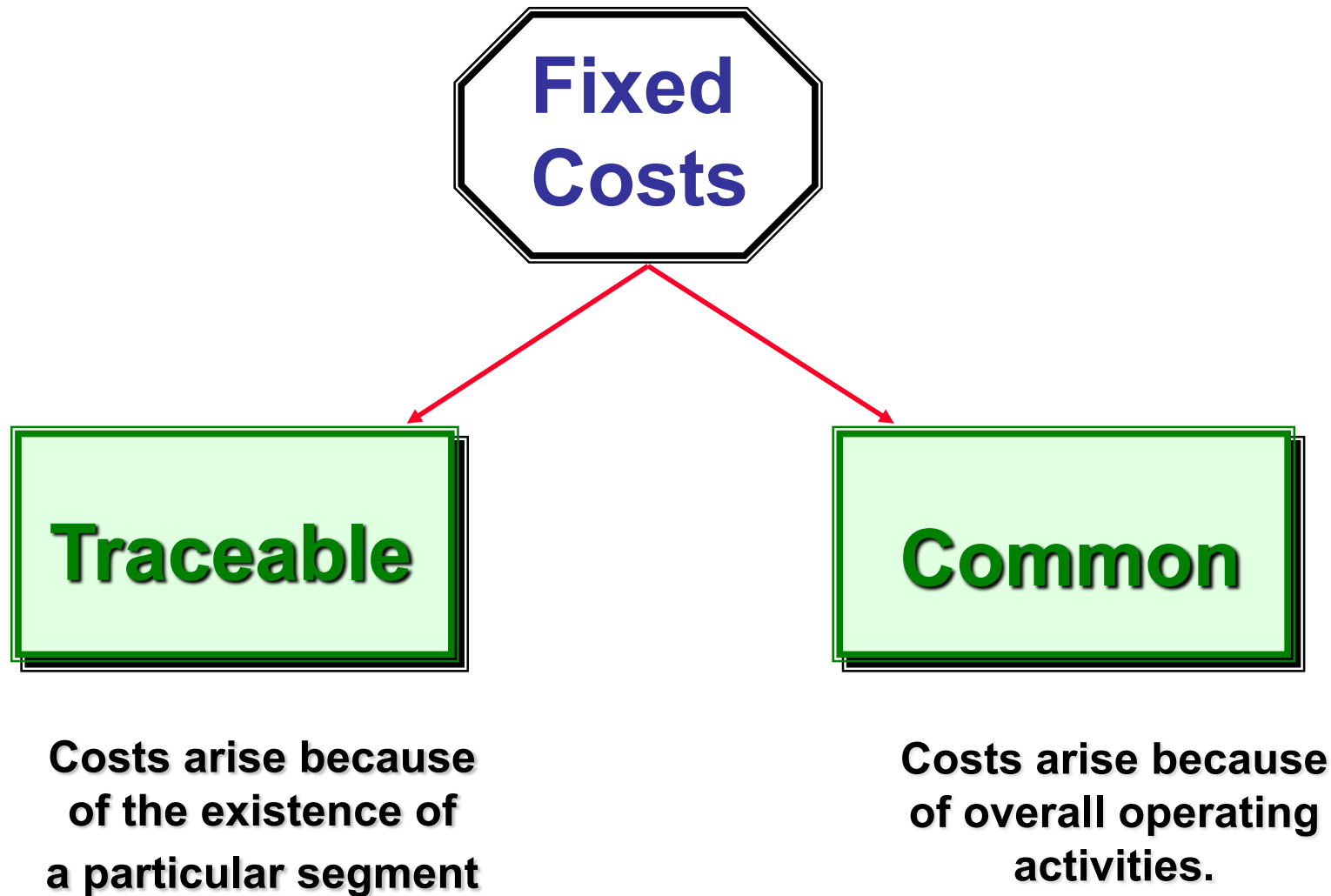
Overheads...

Overhead Confusion...

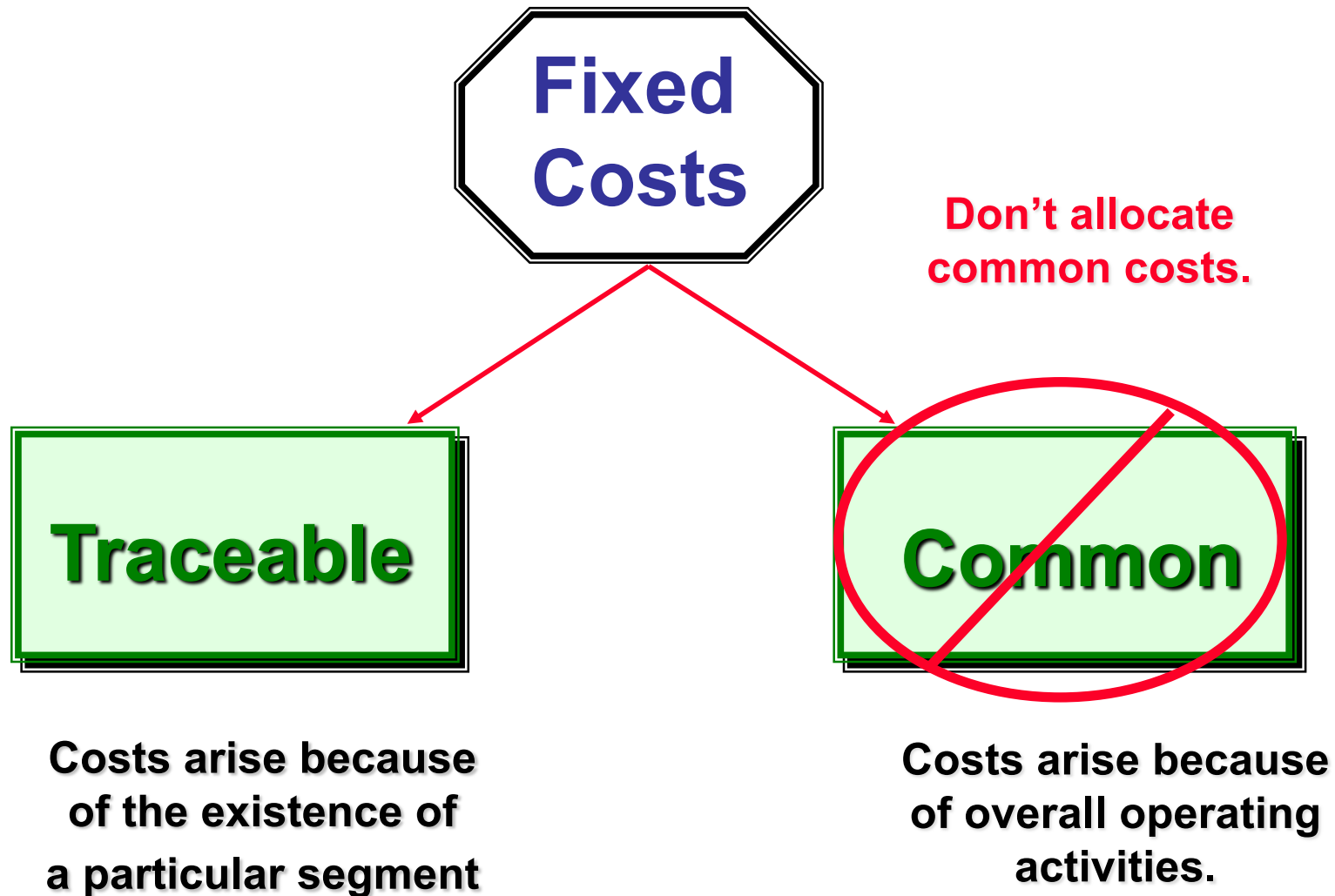
- “The Detroit plant is just a high-cost facility”
- “Where’s the overhead? I mean, we haven’t painted *anything* here in 20 years” (p. 7)

Why is Detroit’s overhead so high?

Traceable and Common Costs



Traceable and Common Costs



Levels of Segmented Statements

Income Statement			
	Company	Product 1	Product 2
Sales	\$ 500,000	\$ 300,000	\$ 200,000
Variable costs	230,000	150,000	80,000
CM	270,000	150,000	120,000
Traceable FC	170,000	90,000	80,000
Division margin	100,000	\$ 60,000	\$ 40,000
Common costs	25,000		
Net operating income	\$ 75,000		

Common costs should not be allocated to the Products. These costs would remain even if one of the Products were eliminated.

Decisions to Eliminate Segments/Products

Internal accounting: Discontinue product as operational costs were high and product not profitable.



Moving Products

- The Study Group's proposal is to move production of Groups 1 and 2 to other factories

“Lack” of capacity at other factories

<i>Type of Product</i>	<i>"Good" On-Highway Axles</i>			<i>"Good" Off-Highway Axles</i>	
<i>Suggestion</i>	<i>Send to Fremont</i>	<i>Send to Maysville</i>	<i>Send to Lancaster</i>	<i>Send to Saginaw</i>	<i>Send to Lima</i>
<i>Group</i>	<i>Group 1</i>	<i>Group 1</i>	<i>Group 1</i>	<i>Group 2</i>	<i>Group 2</i>
Detroit Sales (\$000)	\$13,888	\$13,888	\$13,888	\$9,258	\$9,258
Pct Cap at destination	88%	73%	???	94%	20%
Capacity at destination	\$180,000	\$150,000	\$106,000	\$100,000	\$60,000
Sales at destination	\$158,000	\$110,000	???	\$94,200	\$12,000
\$ Sales of unused capacity at destination	\$22,000	\$40,000	???	\$5,800	\$48,000

Moving Products

- Incentive issues due to how plants are funded and managed

ROI - A Major Drawback

- As division manager at a Plant, your compensation package includes a salary plus bonus based on your division's ROI -- the higher your ROI, the bigger your bonus.
- The company requires an ROI of 20% on all new investments -- your division has been producing an ROI of 30%.
- You have an opportunity to invest in a new project that will produce an ROI of 25%.

As division manager would you invest in this project?

ROI - A Major Drawback

**Gee . . .
I thought we were
supposed to do what
was best for the
company!**

**As division manager,
I wouldn't invest in
that project because
it would lower my pay!**



Manage (TLC)

- Job shops exist in large organizations as they generally add a lot of value.
- This value needs to be recognized clearly and the Job shop needs to be suitably managed.
 - People
 - Process
 - investments

Measure Manage **FUND**



Shouldice
HERNIA HOSPITAL



Shouldice
HERNIA HOSPITAL

Some preliminary observations

	B.C.	Aravind	Difference (Times)
Clinical Labour	491.53	21.23	23.15
Medicine and Surgical Consumables	191.42	21.72	8.81
Linen	19.00	0.16	115.27
Repair & Maintenance (Instruments)	15.00	2.46	6.10
Incremental Resources (?)	50.10	4.16	12.04
Depreciation (Instruments and Equipment)	23.20	1.43	16.25
Total	\$790.25	\$51.16	15.45

Surgeons and Staff: B.C.

	Minutes/Cas e	Cost/Case
OR Booking Clerk	26.00	13.13
Intraoperative RN	30.00	26.54
Pre/Post-Op RN	30.00	26.54
Sedation RN	30.00	26.54
Housekeeping	3.50	1.55
CSD Techs	50.00	24.39
Surgeon	38.67	342.00
		460.68

Surgeons and Staff: Aravind

	Minutes/Cas e	Cost/Case
OR Booking Clerk	5.00	0.071
Intraoperative RN	3.00	0.043
Pre/Post-Op RN	2.00	0.028
Sedation RN	4.00	0.057
Housekeeping	2.63	0.037
CSD Techs	50.00	0.710
Surgeon	5.00	1.250
		2.196